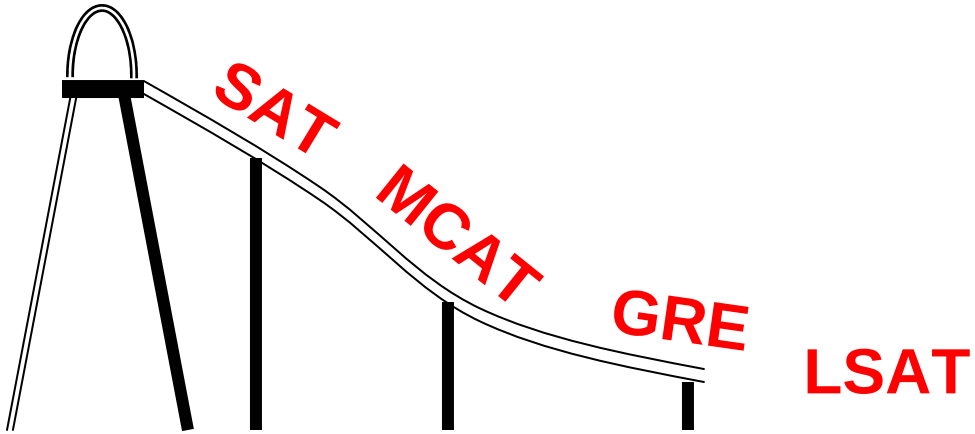
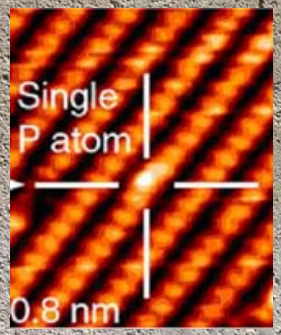
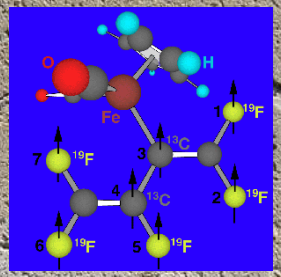
Test Slide

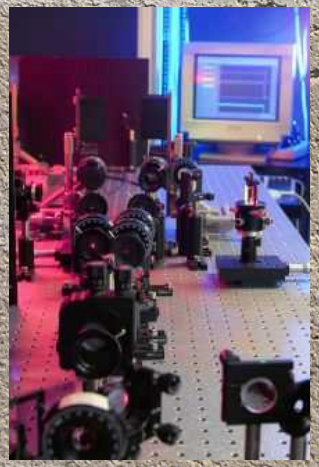


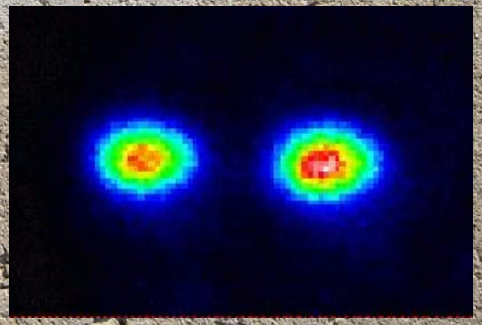
Towards Reliable Quantum Computers

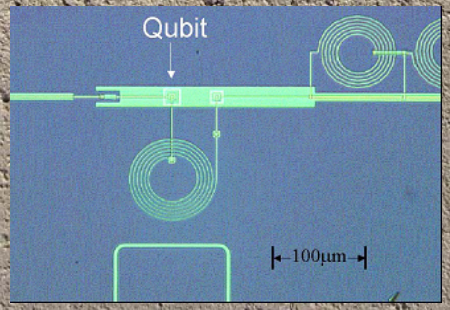


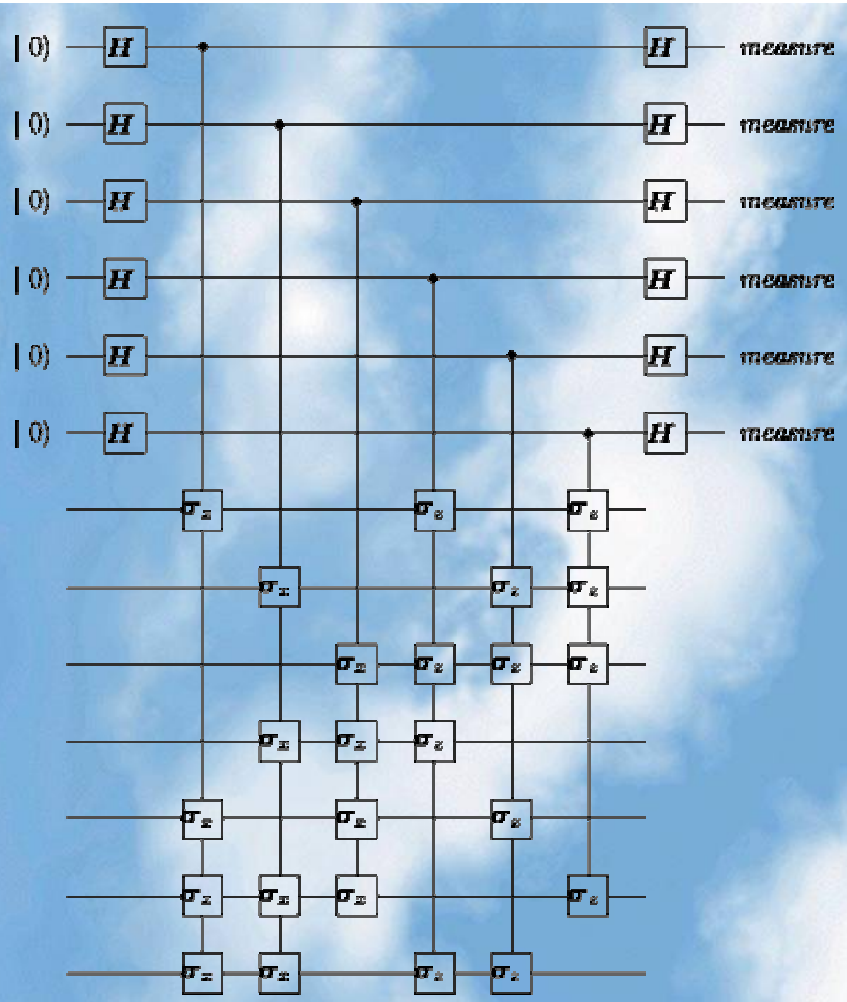












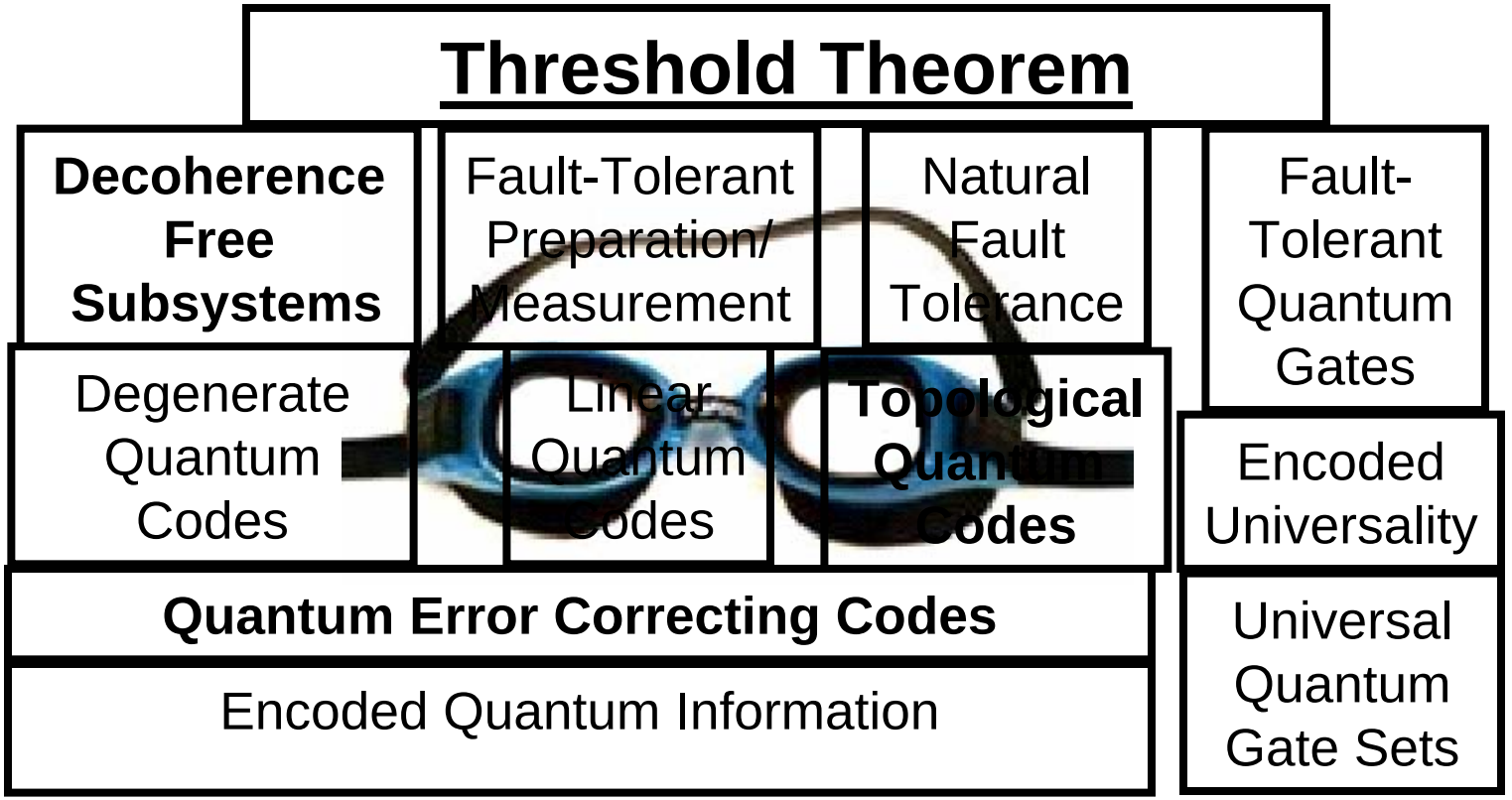
Dave Bacon

IQI

Department of Physics

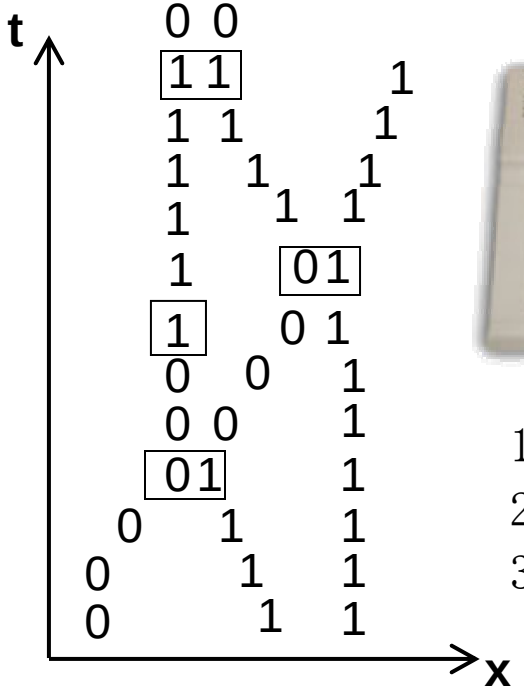
Caltech

Theory Goggles



Quantum Digital?

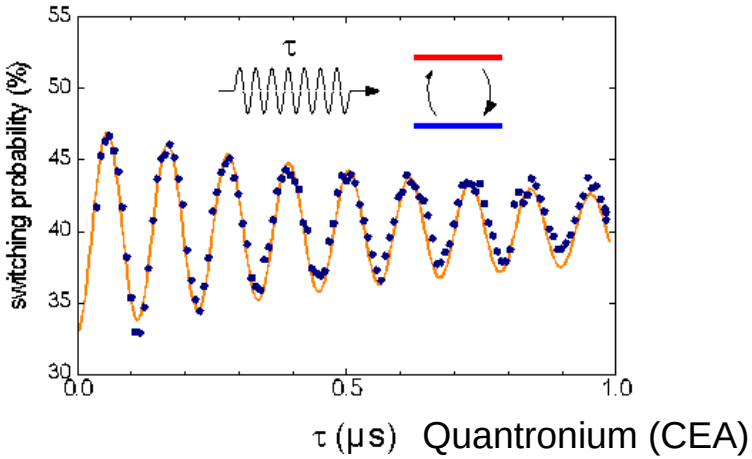
Classical Computer





```
10 PRINT  “BEEP!”
20 PRINT CHR$(7)
30 GOTO 10
```

Quantum Physics Experiment



Quantum Data

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Digital Data

Digital Manipulations

Quantum Classical Transition

Quantum Manipulations

$$\mathbf{U}(t) = \exp\left(-\frac{i\mathbf{H}t}{\hbar}\right)$$

Is a Quantum Computer really a Computer?

Five Problems with Physics *Analogs*

1. quantum data is analog

   properly defined complexity class BQP

   amplitudes like probabilities

2. quantum data likes to become classical data

   quantum error correcting codes

   error avoidance methods

   selection rules, degeneracy, and cooling

3. quantum evolution is analog

   fault-tolerant gate sets

   thermodynamics of refrigeration

   digitizing quantum errors

   linearity of quantum theory

4. quantum preparation is imprecise

   fault-tolerant encoded state preparation

5. quantum measurement is imprecise

   fault-tolerant measurement procedures

Solution Set

Quantum Error Correction

Barbara Terhal (IBM)

Topological Codes

David Meyer (UC San Diego)

Decoherence-Free Methods and Decoupling Techniques

Daniel Lidar (Toronto)

Architectural Aspects of Ion Trap Quantum Computers

Isaac Chuang (MIT)

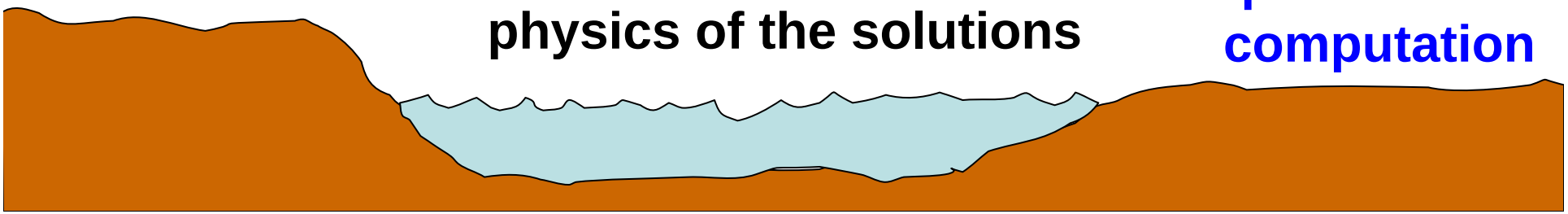
physics: a.k.a. "the real world"

This Talk

the big picture

physics of the solutions

robust quantum computation



Decoherence

2. quantum data likes to become classical data

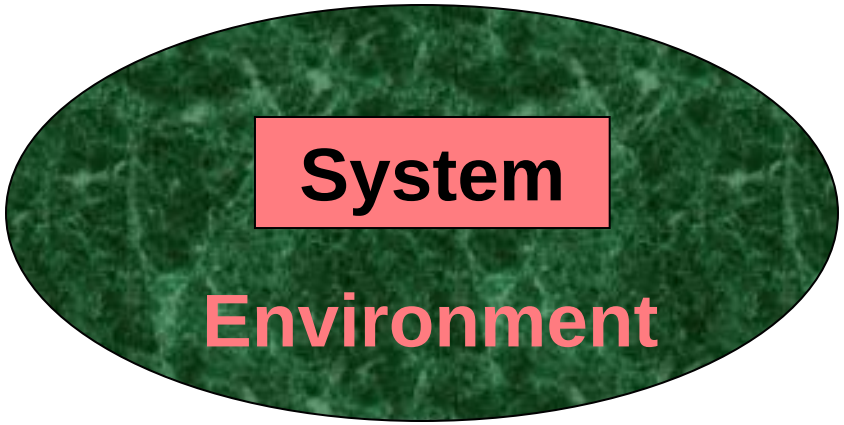
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

amplitudes

open system dynamics

$$p|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|$$

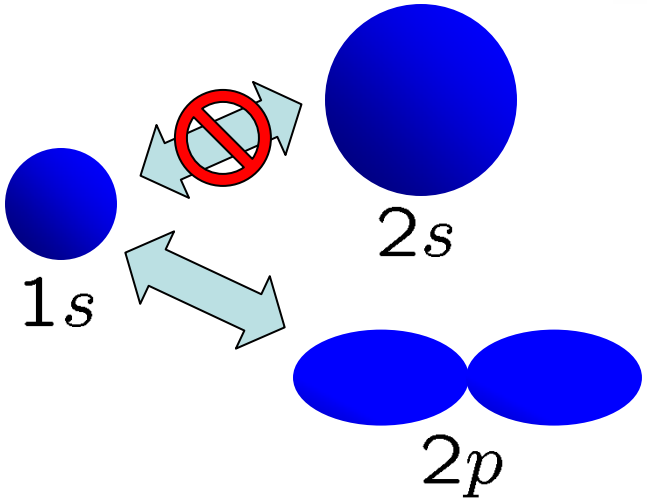
probabilities



How to **guide** the dynamics of system-environment coupling?

Selection Rules

electric dipole interaction: $p_{|i\rangle \to |f\rangle} \propto |\langle f|\hat{\mathbf{r}} \cdot \hat{\boldsymbol{\epsilon}}|i\rangle|^2$



orbital angular momentum selection rule:

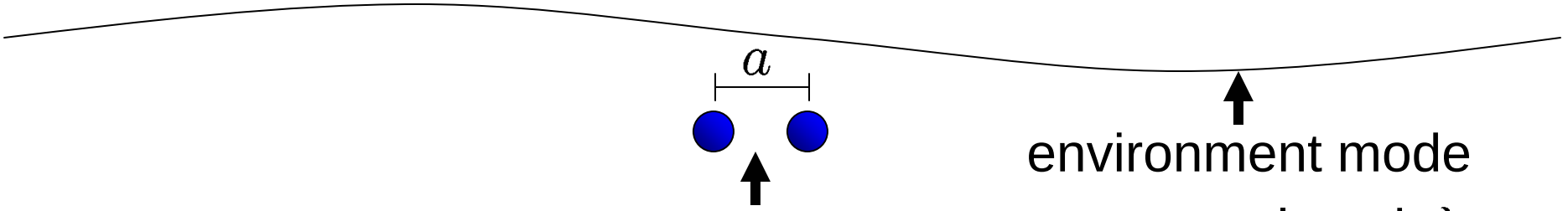
$$\Delta l = \pm 1$$

Different methods differ mostly in how they treat the symmetry of the interaction!

symmetry of system-environment interaction
+
symmetry of states
=
restricted system dynamics due to system-environment interaction

Decoherenc-Free: use natural symmetry of interaction
Decoupling Methods: engineer effective interaction with a symmetry
Quantum Error Correction: use symmetry of noise model

System-Environment Symmetries



$\lambda \gg a$ environment mode cannot distinguish between qubits

symmetry

interaction Hamiltonian

qubit 1 qubit 2 environment

$$\mathbf{H}_I = \sum_\alpha [\mathbf{s}_\alpha \otimes \mathbf{I}] \otimes \mathbf{A}_\alpha + [\mathbf{I} \otimes \mathbf{s}_\alpha] \otimes \mathbf{B}_\alpha$$

symmetry

$$\mathbf{H}_I = \sum_\alpha [\mathbf{s}_\alpha \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{s}_\alpha] \otimes \mathbf{C}_\alpha$$

$s_\alpha,\ \alpha = \{x, y, z\}$

spin ½ Pauli operators

collective decoherence model

Decoherence-Free

$$\mathbf{H}_I = \sum_\alpha [\mathbf{s}_\alpha \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{s}_\alpha] \otimes \mathbf{C}_\alpha = \sum_\alpha \mathbf{S}_\alpha \otimes \mathbf{C}_\alpha$$

total spin operators: $\mathbf{S}_\alpha = \mathbf{s}_\alpha \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{s}_\alpha$

spin ½ → spin 1:

$$|S=1,m=1\rangle = |00\rangle$$
$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$|S=1,m=-1\rangle = |11\rangle$$

spin ½ → spin 0:

$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

add spin ½

$$\mathbf{S}_\alpha |S=0,m=0\rangle = 0 \quad \text{for all } \alpha$$

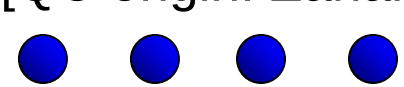
$$\mathbf{H}_I |S=0,m=0\rangle = 0$$

$$\mathbf{H}_I |S=1,m\rangle \neq 0 \quad \langle S=0,m=0|\mathbf{H}_I|S=1,m\rangle = 0$$

singlet does not evolve due to system-environment coupling!

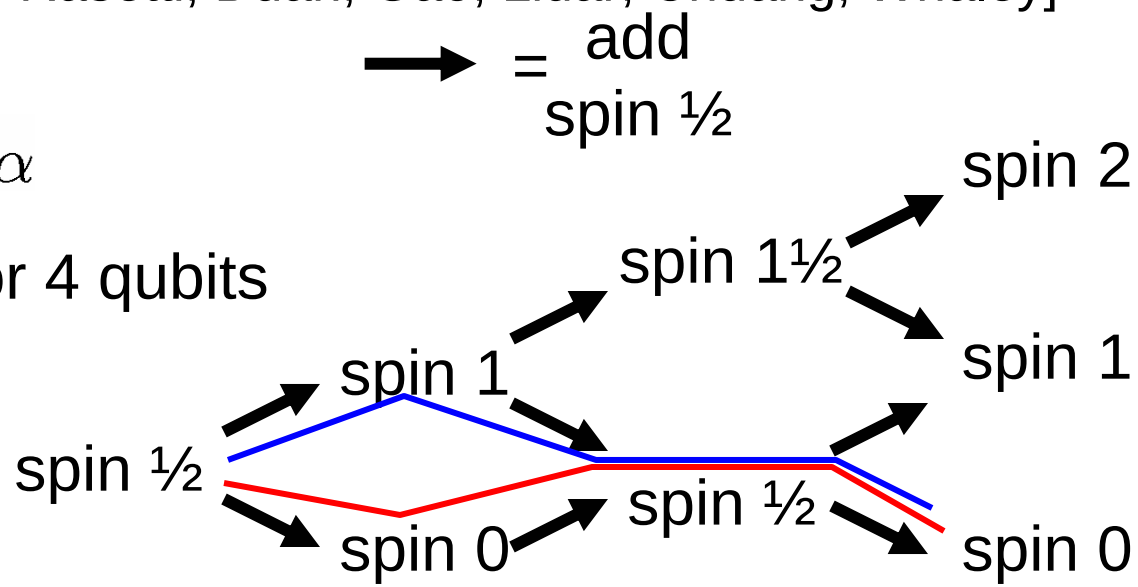
Decoherence-Free Subspaces

[QC origin: Zanardi, Rasetti, Duan, Gao, Lidar, Chuang, Whaley]



$$\mathbf{H}_I = \sum_\alpha \mathbf{S}_\alpha \otimes \mathbf{C}_\alpha$$

total spin operators for 4 qubits



two spin 0 states: $|0_L\rangle = |S = 0, m = 0, p = 0\rangle$

$|1_L\rangle = |S = 0, m = 0, p = 1\rangle$

$$\mathbf{S}_\alpha \left(a|0_L\rangle + b|1_L\rangle\right) = 0 \Rightarrow \mathbf{H}_I(a|0_L\rangle + |1_L\rangle) = 0$$

system-environment interaction does not evolve subspace

Decoherence-Free Subspace: encode quantum information into a protected subspace of system. Protected due to symmetry of system-environment interaction.

Bang and then Bang and then…

[QC origin: Viola, Lloyd, Duan, Gao, see also NMR]

DFS system-environment interaction had symmetry:

What if there is no such symmetry?

Engineer the symmetry

$$\mathbf{H} = \mathbf{H}_S \otimes \mathbf{I}_E + \mathbf{I}_S \otimes \mathbf{H}_E + \mathbf{H}_I$$

control over system evolution

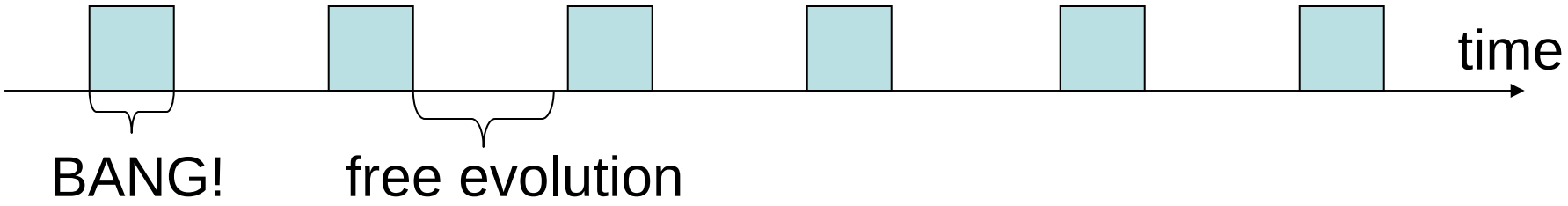
$$\mathbf{U}(t) = \exp(-i\mathbf{H}t)$$

evolution operator

1. $\mathbf{H}_S$ strong compared to $\mathbf{H}_I$

for short enough t $\quad \mathbf{U}(t) \approx \exp(-i\mathbf{H}_S t) \otimes \exp(-i\mathbf{H}_E t)$

$$(\mathbf{U}_1 \otimes \mathbf{V}_1)\mathbf{U}(\Delta t)(\mathbf{U}_2 \otimes \mathbf{V}_2)\mathbf{U}(\Delta t)(\mathbf{U}_3 \otimes \mathbf{V}_3)\cdots$$



Bang and then Bang and then…

2.

$$(\mathbf{U}_1 \otimes \mathbf{V}_1)\exp(-i\mathbf{H}\Delta t)(\mathbf{U}_1^\dagger \otimes \mathbf{W}_1) = \exp(-i\mathbf{H}_{eff}\Delta t)$$

$$\mathbf{H}_{eff} = (\mathbf{U}_1 \otimes \mathbf{V}_1)\mathbf{H}(\mathbf{U}_1^\dagger \otimes \mathbf{W}_1)$$

3.

$$[\exp(-i\mathbf{H}_1\Delta t)\exp(-i\mathbf{H}_2\Delta t)]^{\frac{t}{\Delta t}}$$

$$\approx \exp(-i(\mathbf{H}_1 + \mathbf{H}_2)t)$$

Engineered effective evolution

$$\mathbf{H}_{eff} = \sum_i (\mathbf{U}_i \otimes \mathbf{V}_i)\mathbf{H}(\mathbf{U}_i^\dagger \otimes \mathbf{W}_i)$$

we control: $\mathbf{U}_i$

$$\mathbf{U}(t) \approx \exp(-i\mathbf{H}_{eff}t)$$

We have "symmetrized" the evolution

Decoupling

bare Hamiltonian:

$$\mathbf{H} = \mathbf{s}_z \otimes \mathbf{B}$$

system environment

effective Hamiltonian:

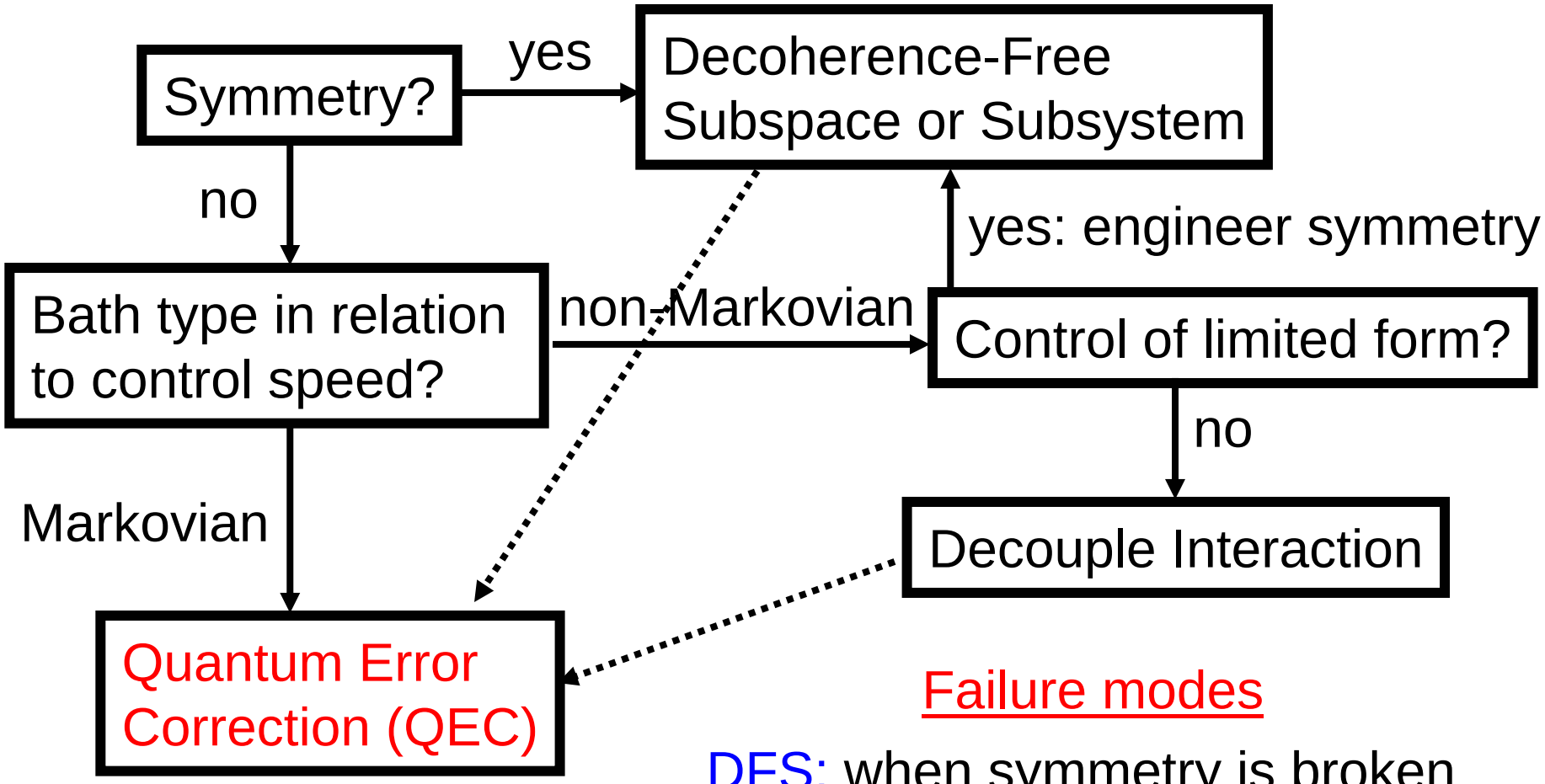
$$\mathbf{H}_{eff} = (\mathbf{X} \otimes \mathbf{V}_1)\mathbf{H}((\mathbf{X}^\dagger \otimes \mathbf{W}_1) + \mathbf{H}$$

$$\mathbf{X}\mathbf{s}_z\mathbf{X}^\dagger = -\mathbf{s}_z \Rightarrow \mathbf{H}_{eff} = \mathbf{0} \qquad \mathbf{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\mathbf{H}_{eff} = \sum_i (\mathbf{U}_i \otimes \mathbf{V}_i)\mathbf{H}(\mathbf{U}_i^\dagger \otimes \mathbf{W}_i)$$

Symmetrized evolutions can eliminate evolution
or
give interaction particular symmetry (DFS)

Go With the Flow

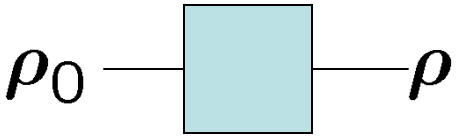


Failure modes

DFS: when symmetry is broken

Decoupling: not fast enough
higher-order interactions

Quantum Error Correcting Symmetry



closed system evolution

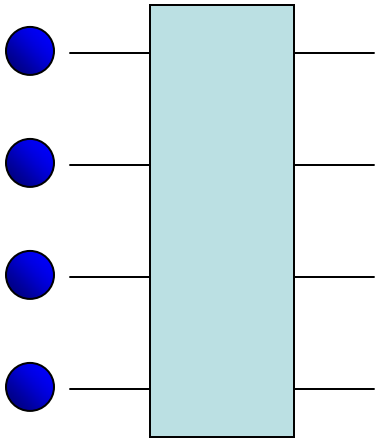
$$\rho = \mathrm{U}\rho_0\mathrm{U}^\dagger$$

$$\mathrm{U}\mathrm{U}^\dagger = \mathrm{I}$$

open system evolution

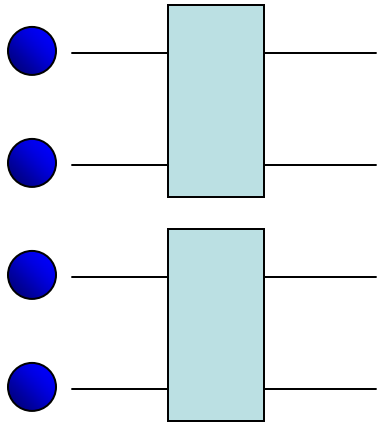
$$\rho = \sum_i \mathrm{A}_i\rho_0\mathrm{A}_i^\dagger$$

$$\sum_i \mathrm{A}_i^\dagger\mathrm{A}_i = \mathrm{I}$$

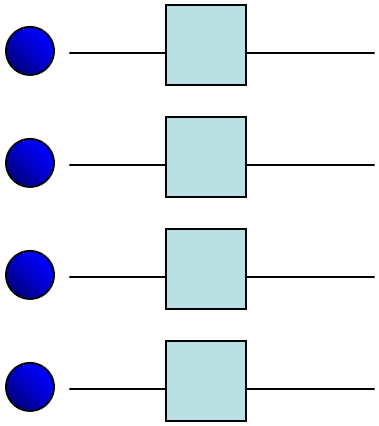


generic errors

no symmetry at all: we are dead!



spatially correlated errors



independent errors

symmetry of single qubit errors

Single Qubit Symmetries

2 qubits



$\mathbf{X}\otimes\mathbf{X}$	$\mathbf{Z}\otimes\mathbf{Z}$	eigenstate
$+1$	$+1$	$\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$
$+1$	-1	$\frac{1}{\sqrt{2}}(\|01\rangle + \|10\rangle)$
-1	$+1$	$\frac{1}{\sqrt{2}}(\|00\rangle - \|11\rangle)$
-1	-1	$\frac{1}{\sqrt{2}}(\|01\rangle - \|10\rangle)$

$$\mathbf{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$\mathbf{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$+1, +1$ eigenstate: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$\mathbf{X}\otimes\mathbf{I}$ → $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

$\mathbf{Y}\otimes\mathbf{I}$ → $\frac{i}{\sqrt{2}}(|01\rangle - |10\rangle)$

$\mathbf{Z}\otimes\mathbf{I}$ → $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

orthogonal to $+1, +1$ eigenstate

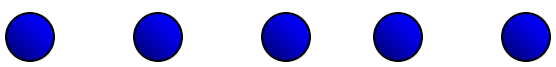
measuring $\mathbf{X}\otimes\mathbf{X}$ and $\mathbf{Z}\otimes\mathbf{Z}$ detects single qubit errors

$+1, +1$ eigenstate: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$$\xrightarrow{\mathbf{E}\otimes\mathbf{I}} \alpha|+1,-1\rangle + \beta|-1,+1\rangle + \gamma|-1,-1\rangle$$

Quantum Error Correcting Code

[Shor, Calderbank, Steane, Lafflame, Knill, Gottesman, Bennett, DiVincenzo, Smolin, Wootters...]



5 qubits

$$\mathbf{X}\otimes\mathbf{Z}\otimes\mathbf{Z}\otimes\mathbf{X}\otimes\mathbf{I}$$

$$\mathbf{I}\otimes\mathbf{X}\otimes\mathbf{Z}\otimes\mathbf{Z}\otimes\mathbf{X}$$

$$\mathbf{X}\otimes\mathbf{I}\otimes\mathbf{X}\otimes\mathbf{Z}\otimes\mathbf{Z}$$

$$\mathbf{Z}\otimes\mathbf{X}\otimes\mathbf{I}\otimes\mathbf{X}\otimes\mathbf{Z}$$

4 commuting operators

simultaneous +1 eigenvalues

NEWNESS

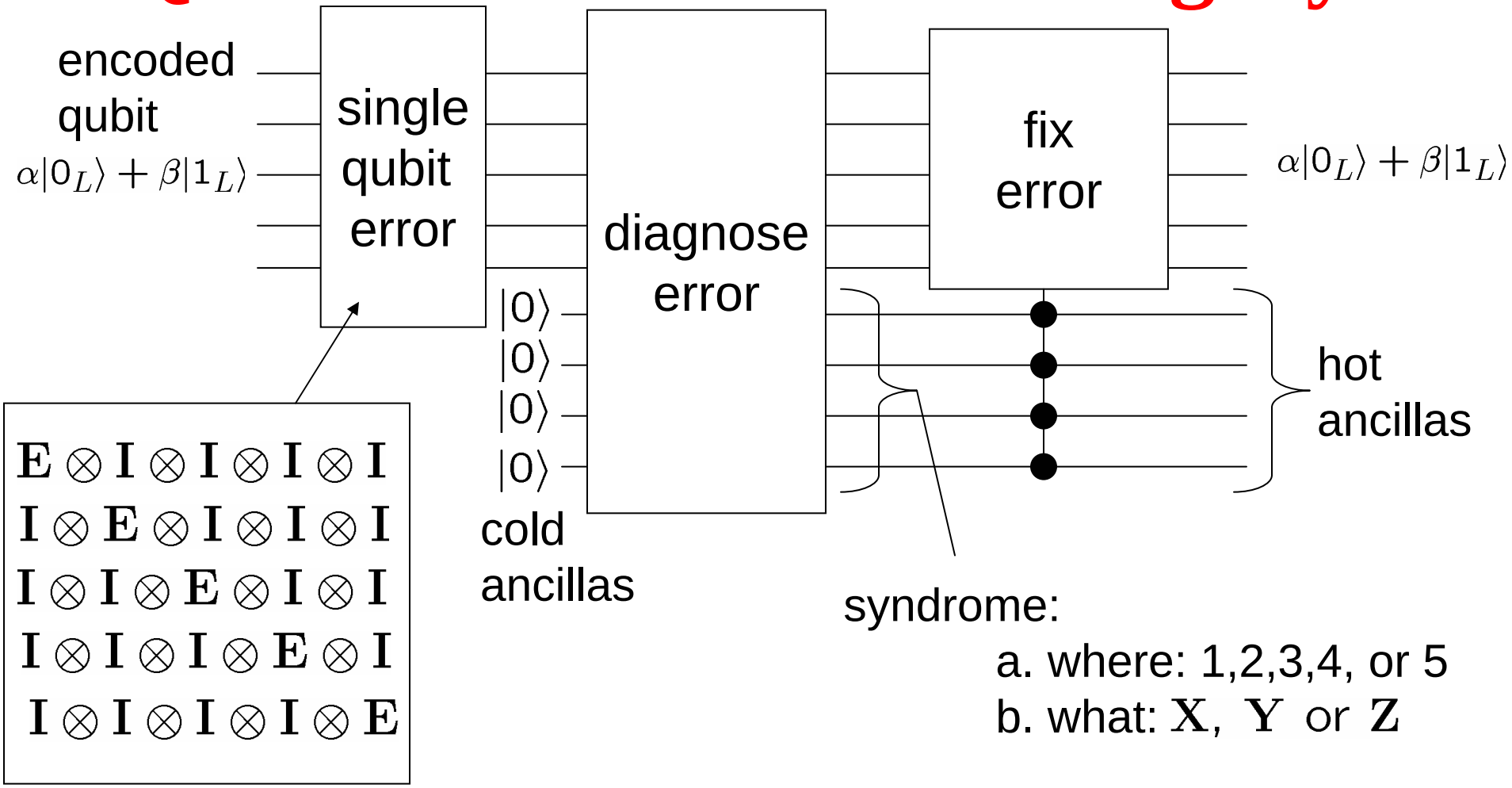
1. encoded qubit $|0_L\rangle,\ |1_L\rangle$

+1 eigenspace is two dimensional

2. not only can we detect single qubit errors
we can detect where any single qubit error occurred

		$\mathbf{X}\otimes\mathbf{I}\otimes\mathbf{I}\otimes\mathbf{I}\otimes\mathbf{I}$	$\mathbf{I}\otimes\mathbf{Z}\otimes\mathbf{I}\otimes\mathbf{I}\otimes\mathbf{I}$
$\mathbf{X}\otimes\mathbf{Z}\otimes\mathbf{Z}\otimes\mathbf{X}\otimes\mathbf{I}$	$+1$	$+1$	$+1$
$\mathbf{I}\otimes\mathbf{X}\otimes\mathbf{Z}\otimes\mathbf{Z}\otimes\mathbf{X}$	$+1$	$+1$	-1
$\mathbf{X}\otimes\mathbf{I}\otimes\mathbf{X}\otimes\mathbf{Z}\otimes\mathbf{Z}$	$+1$	$+1$	$+1$
$\mathbf{Z}\otimes\mathbf{X}\otimes\mathbf{I}\otimes\mathbf{X}\otimes\mathbf{Z}$	$+1$	-1	-1

Quantum Error Correcting Cycle

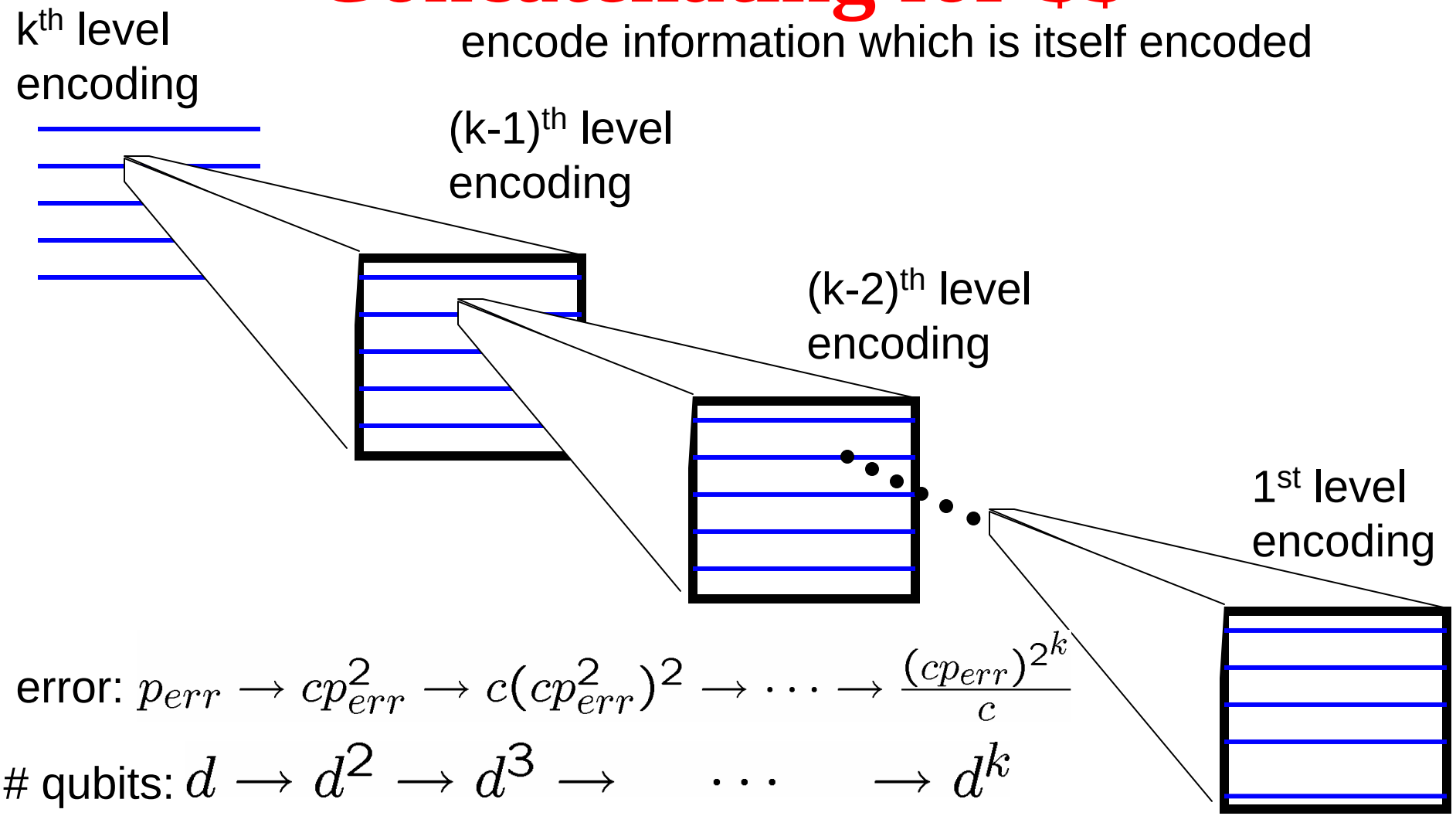


probability of failure unencoded → $p_{err} \rightarrow cp_{err}^2$ ← probability of failure encoded (two qubit errors)

if $p_{err} < \frac{1}{c}$ improvement: $cp_{err}^2 < p_{err}$

Concatenating for $$

encode information which is itself encoded



error: $p_{err} \rightarrow cp_{err}^2 \rightarrow c(cp_{err}^2)^2 \rightarrow \cdots \rightarrow \frac{(cp_{err})^{2^k}}{c}$

qubits: $d \rightarrow d^2 \rightarrow d^3 \rightarrow \quad \cdots \quad \rightarrow d^k$

ultimate accuracy ϵ using $d^k = O(\mathrm{poly}(\log \epsilon^{-1}))$ qubits

Fault-Tolerance

everything can fail:

unitary evolution used for error recovery

preparation of initial state

measurements (wrong result, wrong evolution)

loss of physical qubit

Example of fault-tolerance considerations:

$$|0_L\rangle = |00\cdots0\rangle \quad |1_L\rangle = |11\cdots1\rangle$$

Want gate to flip states: $\mathbf{U}|0_L\rangle = |1_L\rangle \;\; \mathbf{U}|1_L\rangle = |0_L\rangle$

$$\mathbf{U(t)} = \exp(-i\mathbf{X}\otimes\cdots\otimes\mathbf{X}t) = \cos t\mathbf{I} - i\sin t\mathbf{X}\otimes\cdots\otimes\mathbf{X}$$

$$t = \tfrac{\pi}{2} + \epsilon \qquad [-i\mathbf{X}\otimes\cdots\otimes\mathbf{X}](\cos\epsilon\mathbf{I} - i\sin\epsilon\mathbf{X}\otimes\cdots\mathbf{X})$$

extra evolution not fixable

$$\mathbf{U(t)} = \exp(-i(\mathbf{X}_1+\mathbf{X}_2+\cdots+\mathbf{X}_n)t) = \mathbf{U}_s(t)\otimes\mathbf{U}_s(t)\otimes\cdots\otimes\mathbf{U}_s(t)$$

$$\mathbf{U}_s(t) = \cos t\mathbf{I} - i\sin t\mathbf{X}$$

$$t = \tfrac{\pi}{2} + \epsilon \quad [(-i)^n\mathbf{X}\otimes\cdots\otimes\mathbf{X}]\mathbf{U}_s(\epsilon)\otimes\cdots\otimes\mathbf{U}_s(\epsilon)$$

extra evolution like single qubit errors!

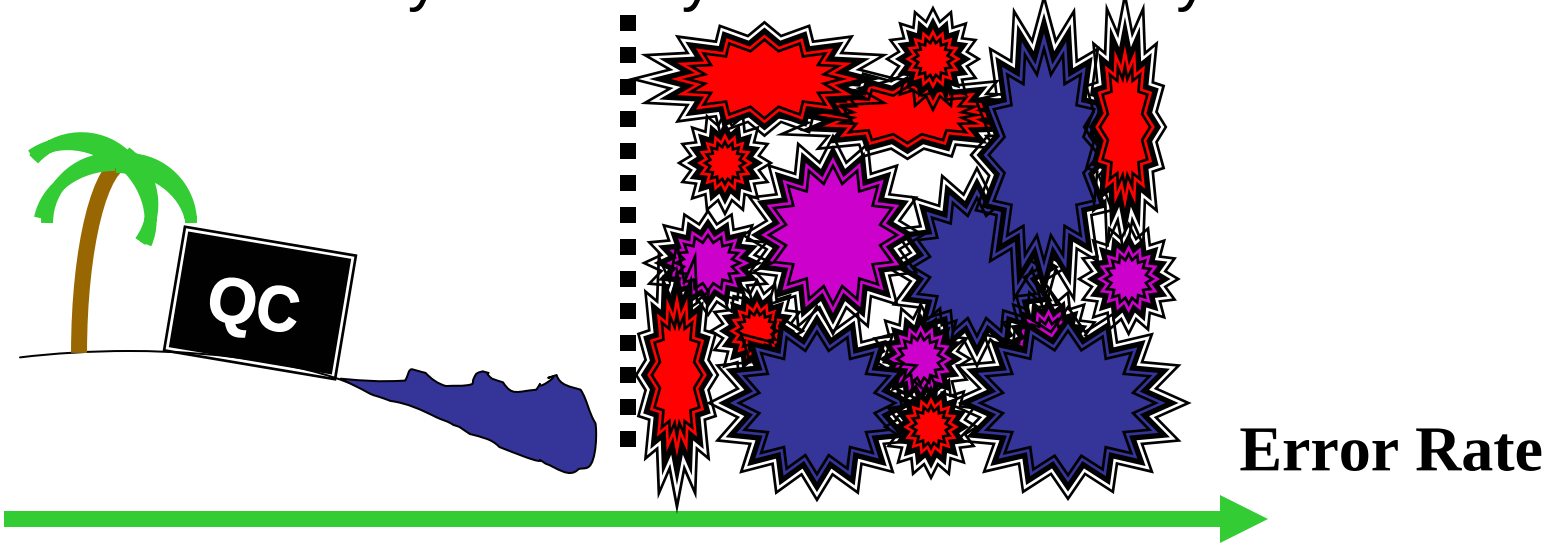
Threshold Theorem

Fault Tolerant Quantum Computing is Possible!

[Shor, Gottesman, Preskill, Aharonov, Ben-Or, Kitaev, Knill, Laflamme, Zurek, Terhal, Burkard, …]

Threshold Theorem:

If the error rate (typical number of operations per decoherence time/ accuracy in quantum evolution / measurement error rate) is smaller than a certain value (the threshold), then quantum computation to arbitrary accuracy can be efficiently enacted.

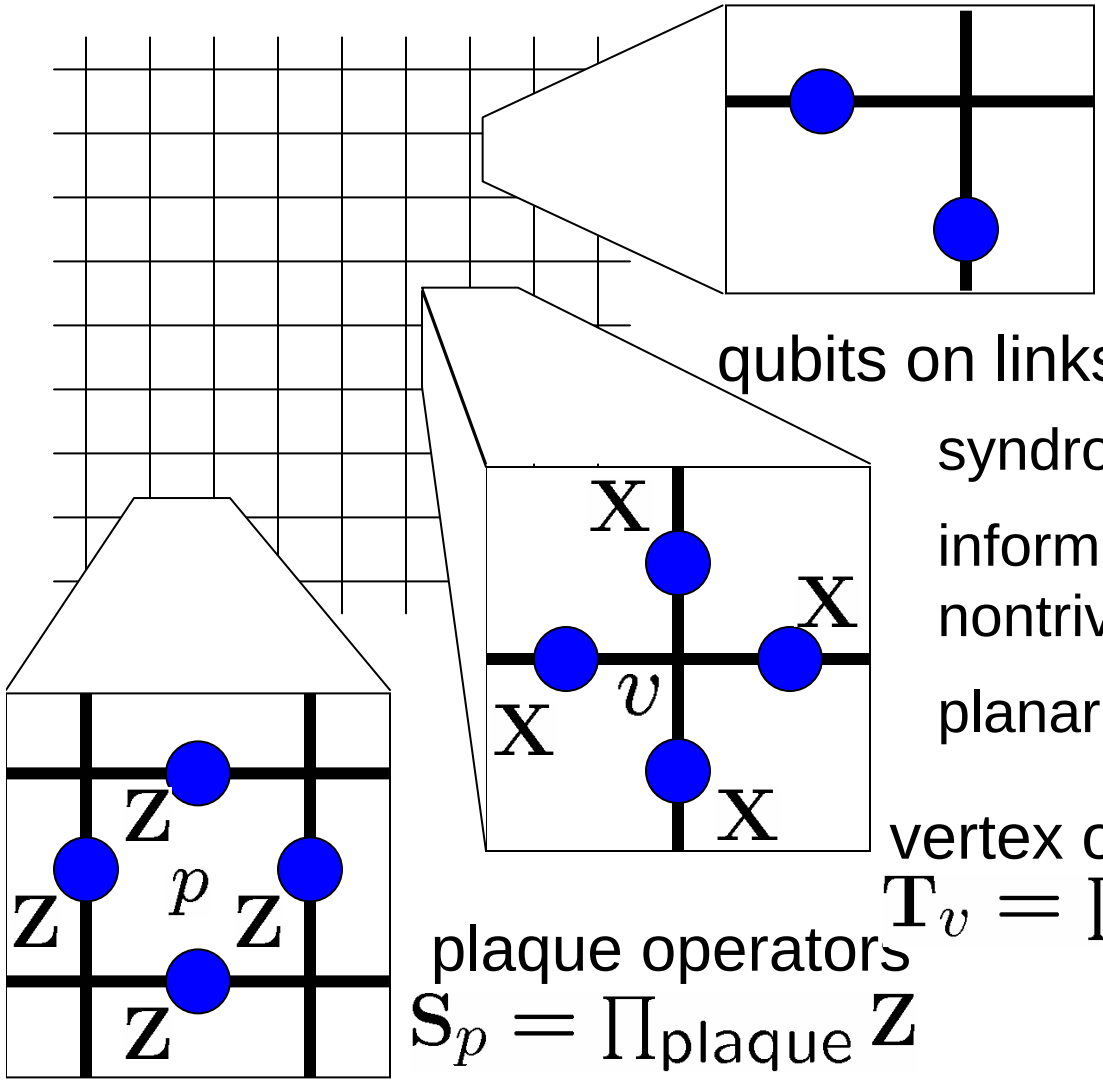


Quantum Computation is NOT Analog Computation

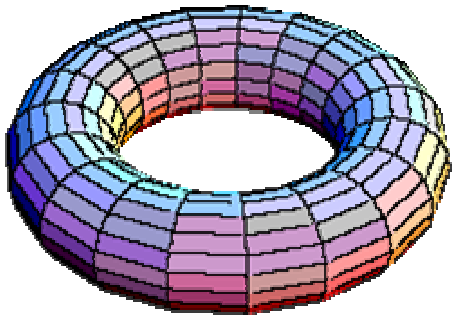
Topological Quantum Codes

[Kitaev, Preskill, Freedman, Meyer,…]

Concatentation is hierarchical solution to Threshold



Kitaev's toric code



qubits on links

syndrome measurements local

information in topologically nontrivial operators

planar codes also (torus not needed)

vertex operators

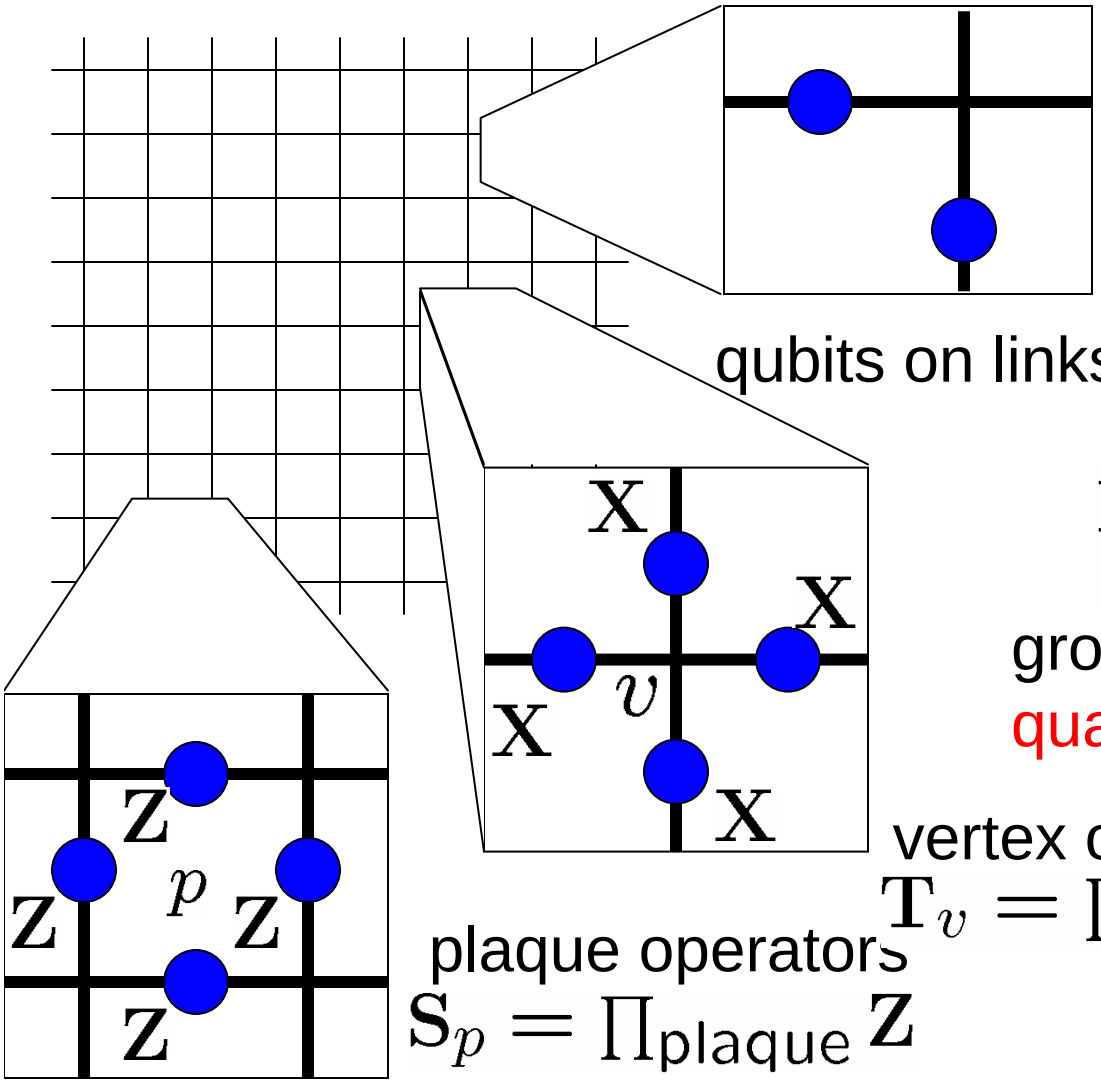
$$\mathbf{T}_v = \prod_{\text{vertex}} \mathbf{X}$$

plaque operators

$$\mathbf{S}_p = \prod_{\text{plaque}} \mathbf{Z}$$

Natural Quantum Error Correction

[Kitaev]



qubits on links

$$\mathbf{H} = -\sum_p \mathbf{S}_p - \sum_v \mathbf{S}_v$$

ground state is gapped and is a quantum error correcting code

vertex operators

$$\mathbf{T}_v = \Pi_{\text{vertex}}\, \mathbf{X}$$

plaque operators

$$\mathbf{S}_p = \Pi_{\text{plaque}}\, \mathbf{Z}$$

Quantum error correction done naturally by the physics?

Rant mode ON

The Physics Guarantee

There are distinct PHYSICAL and DYNAMICAL reasons why robust classical computation is possible.





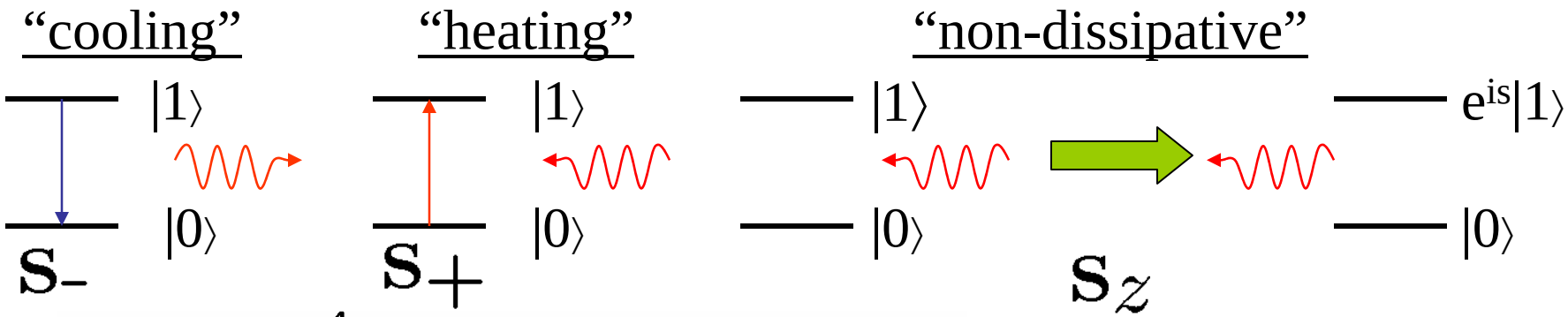
not all physical systems are equally good for computation:

there exist systems whose PHYSICS guarantees their ability to enact robust classical computation.

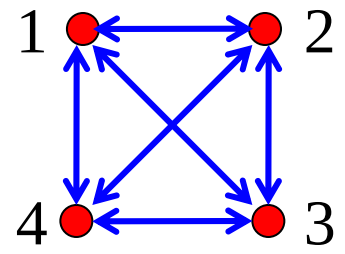
What is the phase of matter corresponding to the quantum computer?

Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

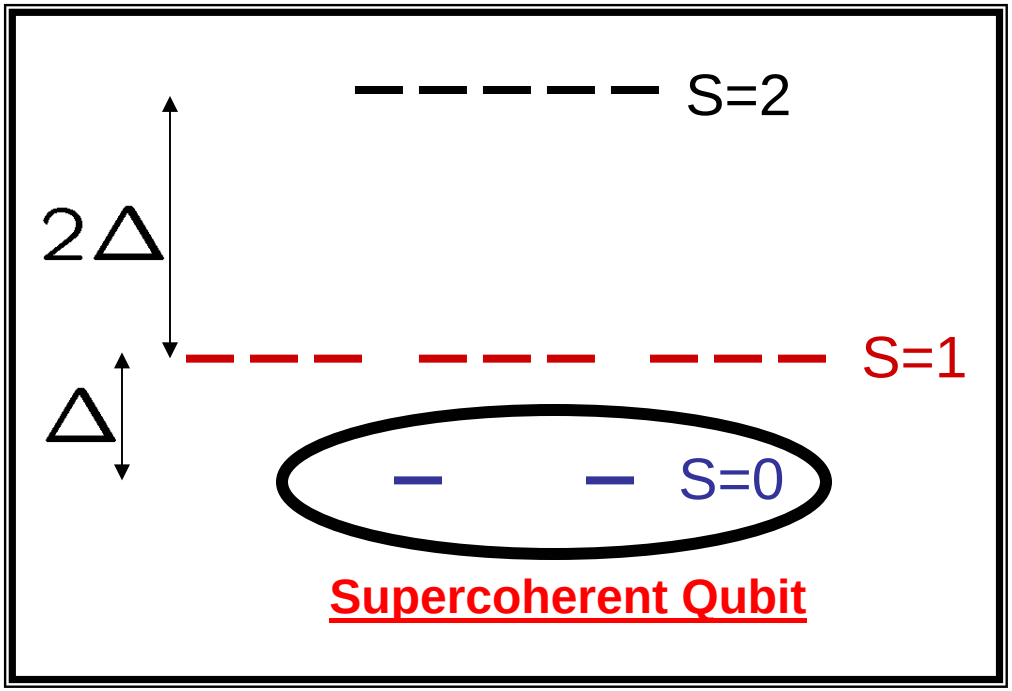
Example: Supercoherence



$$\mathbf{H} = \Delta \sum_{i>j=1}^{4} \hat{\mathbf{s}}_i \cdot \hat{\mathbf{s}}_j = \frac{\Delta}{2}\left(\hat{\mathbf{S}}^2 - 3\mathbf{I}\right)$$



= exchange interaction



ground state is quantum error detecting code

all single qubit errors become heating type errors!

Attack of the Theorists

Quantum Error Correction

Barbara Terhal (IBM)

Topological Codes

David Meyer (UC San Diego)

Decoherence-Free Methods and Decoupling Techniques

Daniel Lidar (Toronto)

Architectural Aspects of Ion Trap Quantum Computers

Isaac Chuang (MIT)



physics: a.k.a. "the real world"

robust quantum computation